THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE UNIT TEST SET A

Chapters 1–4: **Exploration: Entering the World of Chemistry, Exploring Mixtures and Their Separation, The Atomic Foundations of Matter, Journey Inside the Atom** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–12 are Objective Type (1 mark each).
3. Q. 13–17 are Very Short Answer Type (2 marks each).
4. Q. 18–23 are Short Answer Type (3 marks each).
5. Q. 24–25 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 22 to 25.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 12 × 1 = 12

Choose the correct option and write it in your answer book.

1. When a prediction does not match the observation, scientists: 1

(a) re-examine assumptions, models or measurements
(b) stop the study
(c) ignore the observation
(d) replace evidence with belief

2. In chemistry, atoms and molecules are commonly drawn as: 1

(a) spheres joined by bonds **(b)** flow charts
(c) single points **(d)** layers of an onion

3. An aircraft ran short of fuel because the ground crew mixed up: 1

(a) seconds and minutes **(b)** litres and gallons only
(c) pounds and kilograms **(d)** metres and feet

4. A homogeneous mixture of two or more substances with uniform composition throughout is called a: 1

(a) precipitate **(b)** suspension
(c) solution **(d)** heterogeneous mixture

5. Which of these shows the Tyndall effect? 1

(a) sugar solution **(b)** clear vinegar
(c) salt solution **(d)** milk diluted with water

6. Dry ice is: 1

(a) frozen water **(b)** solid carbon dioxide that sublimes
(c) a compound of oxygen **(d)** solid nitrogen that melts

7. The correct name of the covalent compound N_2O_4 is: 1

(a) nitrogen oxide **(b)** nitrogen tetroxide(II)
(c) dinitrogen tetroxide **(d)** dinitrogen tetraoxide

8. 12 g of carbon burns in 32 g of oxygen. The mass of carbon dioxide formed is: 1

(a) 44 g **(b)** 20 g **(c)** 32 g **(d)** 12 g

9. The molecular/formula unit mass of sodium chloride, $NaCl$ (formula unit mass) is (H = 1, C = 12, N = 14, O = 16, Na = 23, Mg = 24, S = 32, Cl = 35.5, Ca = 40): 1

(a) 60.5 u **(b)** 56.5 u **(c)** 58.5 u **(d)** 117 u

10. The valency of aluminium (Z = 13) is: 1

(a) 0 **(b)** 4 **(c)** 3 **(d)** 2

11. The mass number of an atom is the sum of its: 1

(a) protons and electrons **(b)** protons and neutrons
(c) electrons and neutrons **(d)** shells and electrons

12. The electronic configuration (K, L, M, N shells) of the potassium atom (Z = 19) is: 1

(a) 2, 8, 9, 0 **(b)** 2, 9, 8 **(c)** 2, 8, 8, 1 **(d)** 2, 8, 8, 2

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 10–15 words each.

13. Why is ignoring some details in a model not a mistake? 2

14. 5 g of glucose is dissolved in water to make 100 mL of solution. Find its % m/v. 2

15. How can acetone be separated from a mixture of acetone and water? Give the principle. 2

16. 4.0 g $CaCO_3$ reacts with 2.92 g HCl to give 1.76 g CO_2, 0.72 g H_2O and 4.44 g $CaCl_2$. Verify the Law of Conservation of Mass. 2

17. Why did Thomson conclude that electrons are present in all atoms? 2

SECTION — C (Short Answer Type) — 6 × 3 = 18

Answer in about 30–40 words each.

18. A message says ‘Do not eat food during an eclipse, it becomes harmful’. How would you check this claim scientifically? 3

19. Why should a hot saturated solution of copper sulfate be filtered before cooling? Why is a drop of dilute sulfuric acid added? 3

20. Explain the Law of Conservation of Mass with the help of the reaction between sodium sulfate and barium chloride solutions. 3

21. Why was Thomson's model rejected after the gold foil experiment? 3

22. State the main features of Rutherford's nuclear (planetary) model. 3

OR

How many atoms of diameter 10^{-10} m must be stacked to make a sheet 0.1 mm thick? 3

23. ‘No scientific theory is ever final.’ Explain why this is a strength of science. 3

OR

Explain, using the aircraft fuel incident, why standard SI units are important. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

24. (i) Distinguish between a solution, a colloid and a suspension with examples. 5
(ii) State the Tyndall effect.
(iii) How would you show that milk is a colloid?

OR

(i) What is paper chromatography? 5
(ii) Describe the experiment to separate the pigments in black ink.
(iii) What would you observe and why?
(iv) Why must the solvent level be below the spot?

25. **Case study:** A hydrogen atom has one proton and a helium atom has two protons, yet helium's mass is four times that of hydrogen. 5
(i) What puzzle does this pose?
(ii) Who resolved it and how?
(iii) What are the nucleons?
(iv) Why do heavier atoms contain many more neutrons than protons?

OR

Case study: An atom X has atomic number 17 and mass number 35. 5
(i) Find protons, neutrons and electrons.
(ii) Write its electronic configuration.
(iii) Find its valency and name the element.
(iv) Predict whether it forms a cation or anion.

END OF QUESTION PAPER — Total: 50 Marks